A.7 Entropy of Answers

Domain	Model	R-Tuning-R	Answer Rate	Min-loss	Answer Rate
ID	OpenLLaMA-3B	82.79	44.65	91.83	26.52
	LLaMA-7B	85.95	44.11	85.78	41.57
	LLaMA-13B	87.06	44.00	88.00	48.33
OOD	OpenLLaMA-3B	44.04	40.80	59.66	20.92
	LLaMA-7B	69.54	28.07	55.52	49.15
	LLaMA-13B	76.13	30.16	60.42	55.75

Table 12: Accuracy (%) and answer rate (%) of R-Tuning-R and min-loss training on ParaRel dataset. The loss is calculated by the first token of the ground-truth answer. ID and OOD denote in-domain and out-of-domain, respectively.

In addition to evaluating the difference between certain and uncertain questions with pre-trained models, we further leverage GPT (Brown et al., 2020) to investigate the patterns of certain and uncertain questions. Specifically, we query `gpt-3.5-turbo` five times with Chain-of-Thought prompts (Wei et al., 2023a) with a temperature of 0.7, and calculate the entropy of the answers toward the same question (Diao et al., 2023b). If the model provides many different answers to the same question, the entropy should be high. Otherwise, the entropy should be low. The results are shown in Table 11. We observe that the average entropy of the answers on certain data D_1 is lower than the entropy of uncertain data D_0 data in most cases, which illustrates that when fed with certain questions, `gpt-3.5-turbo` is more likely to generate consistent answers. It will generate hallucinated answers to uncertain questions with much higher chances.

Therefore, we can conclude that R-Tuning divides the data into two folds. The uncertain questions are generally more difficult than certain questions because `gpt-3.5-turbo`'s answers vary more with the uncertain data. R-Tuning endows the model with abilities to identify and differentiate the difficulties of the questions. Therefore, our fine-tuned model becomes proactive in answering easy questions with certainty while being conservative in answering difficult questions, which eventually increases the precision and prevents the fine-tuned model from making too many mistakes.

A.8 Min-Loss Training

Dataset	Model	R-Tuning	Vanilla
ParaRel	OpenLLaMA-3B	69.79	69.62
	LLaMA-7B	77.45	77.91
	LLaMA-13B	77.69	72.67
MMLU	OpenLLaMA-3B	24.38	24.39
	LLaMA-7B	54.19	63.88
	LLaMA-13B	73.81	74.95
WiCE	OpenLLaMA-3B	56.74	61.05
	LLaMA-7B	55.02	65.47
	LLaMA-13B	71.12	67.17
HotpotQA	OpenLLaMA-3B	46.54	36.90
	LLaMA-7B	57.57	41.92
	LLaMA-13B	57.99	44.76
FEVER	OpenLLaMA-3B	94.22	85.38
	LLaMA-7B	93.30	88.24
	LLaMA-13B	95.23	94.99
HaluEval-QA	OpenLLaMA-3B	73.85	72.11
	LLaMA-7B	77.17	76.22
	LLaMA-13B	80.36	75.73
Average	OpenLLaMA-3B	**61.09**	58.24
	LLaMA-7B	**69.11**	68.94
	LLaMA-13B	**76.03**	71.71

Table 13: Multi-task experiments of R-Tuning and Vanilla with AP scores (%). Vanilla adopts the confidence of the predicted answer to rank the result, while R-Tuning adopts the combination of the confidence of the predicted answer and the confidence of certainty.

Compared with the append verbalizer, replace verbalizer (e.g., R-Tuning-R) is a clear-cut way of producing uncertainty expressions by throwing away valuable labels which could potentially be leveraged for training. In address of this dimension of concern, we consider a modified cross entropy learning objective that pushes up the correct answer token and keeps the uncertainty expressions as the second most probable token choice. We call it min-loss training, which is optimized by gradient descent over the min loss between guessing the correct answer or just uncertainty expressions. It is formulated as follows:

$$\min(L(predict, GT), L(predict, IDK)), \quad (12)$$

where L denotes the cross-entropy loss. To do so, we split the training data in half and adopt a two-stage training strategy. In the first stage, we train our model using the original method where the prompt template uses `The answer is {ground-truth}` if the model answers correctly, otherwise `The answer is unknown`. Once the model learns such a pattern after the first

Dataset	Domain	Model	R-Tuning-U acc.	R-Tuning-U conf.	Vanilla-C acc.	Vanilla-C conf.
ParaRel	ID	OpenLLaMA-3B	61.57	61.50	59.81	75.01
		LLaMA-7B	65.26	71.25	62.68	77.35
		LLaMA-13B	70.47	77.15	65.76	78.50
	OOD	OpenLLaMA-3B	38.39	43.88	37.66	61.56
		LLaMA-7B	42.97	56.00	42.12	63.41
		LLaMA-13B	48.40	61.25	40.77	61.20
MMLU	ID	OpenLLaMA-3B	23.25	44.33	25.54	46.23
		LLaMA-7B	41.86	51.06	34.65	53.64
		LLaMA-13B	41.90	53.87	39.57	58.60
	OOD	OpenLLaMA-3B	25.01	41.97	23.81	44.77
		LLaMA-7B	45.88	55.22	42.66	58.52
		LLaMA-13B	51.74	59.45	50.38	65.53

Table 14: Performance of R-Tuning-U compared with Vanilla-C on the ParaRel and MMLU datasets. ID and OOD denote in-domain and out-of-domain, respectively.

Dataset	Domain	Model	R-Tuning-U	Vanilla-C
ParaRel	ID	OpenLLaMA-3B	49.96	49.93
		LLaMA-7B	50.02	48.80
		LLaMA-13B	59.56	59.28
	OOD	OpenLLaMA-3B	52.26	52.16
		LLaMA-7B	49.45	48.52
		LLaMA-13B	61.21	61.10
MMLU	ID	OpenLLaMA-3B	45.01	44.88
		LLaMA-7B	50.97	50.95
		LLaMA-13B	44.65	43.12
	OOD	OpenLLaMA-3B	43.10	42.98
		LLaMA-7B	58.34	58.33
		LLaMA-13B	64.13	62.37

Table 15: The average sureness probability (%) of R-Tuning-U and Vanilla-C.

Dataset	Domain	Model	R-Tuning-U	Vanilla-C
ParaRel	ID	OpenLLaMA-3B	0.018	0.255
		LLaMA-7B	0.057	0.250
		LLaMA-13B	0.064	0.228
	OOD	OpenLLaMA-3B	0.054	0.291
		LLaMA-7B	0.132	0.271
		LLaMA-13B	0.124	0.258
MMLU	ID	OpenLLaMA-3B	0.212	0.246
		LLaMA-7B	0.092	0.243
		LLaMA-13B	0.120	0.239
	OOD	OpenLLaMA-3B	0.172	0.258
		LLaMA-7B	0.093	0.209
		LLaMA-13B	0.078	0.200

Table 16: The ECE (Expected Calibration Error) of R-Tuning-U and Vanilla-C.

training stage, we calculate the min-loss with the equation 12. We only consider the loss of the unknown and the ground-truth label, and we mask the tokens before them. Since the ground-truth label may consider more than one token, we calculate the loss for the first token.

We evaluate the performance of min-loss strategy on the ParaRel dataset, and the results are shown in Table 12. It shows min-loss training outperforms R-Tuning-R in small models and in-domain settings. However, it underperforms R-Tuning-R in out-of-domain test sets. We also notice that in out-of-domain test sets, the accuracy of the model of 3B size is nearly the same as 7B's and 13B's, but its answer rate is much lower. We identify such issues as a trade-off between the accuracy and the answer rate. When the model is proactive in answering more questions, it will inevitably make more mistakes. As the intrinsic parametric knowledge of the model is limited, there is no method to fine-tune a model with both high accuracy and a high answer rate.

A.9 Confidence Distribution of Training Dataset

We calculate the confidence of the certain data D_1 and uncertain data d_0, and they are shown in Figure 7.

A.10 AP Scores of Each Dataset and Model Size with Figures

We calculate the AP scores for each dataset with different model sizes in multi-task experiments. The results are shown in Table 13 and Figure 8.

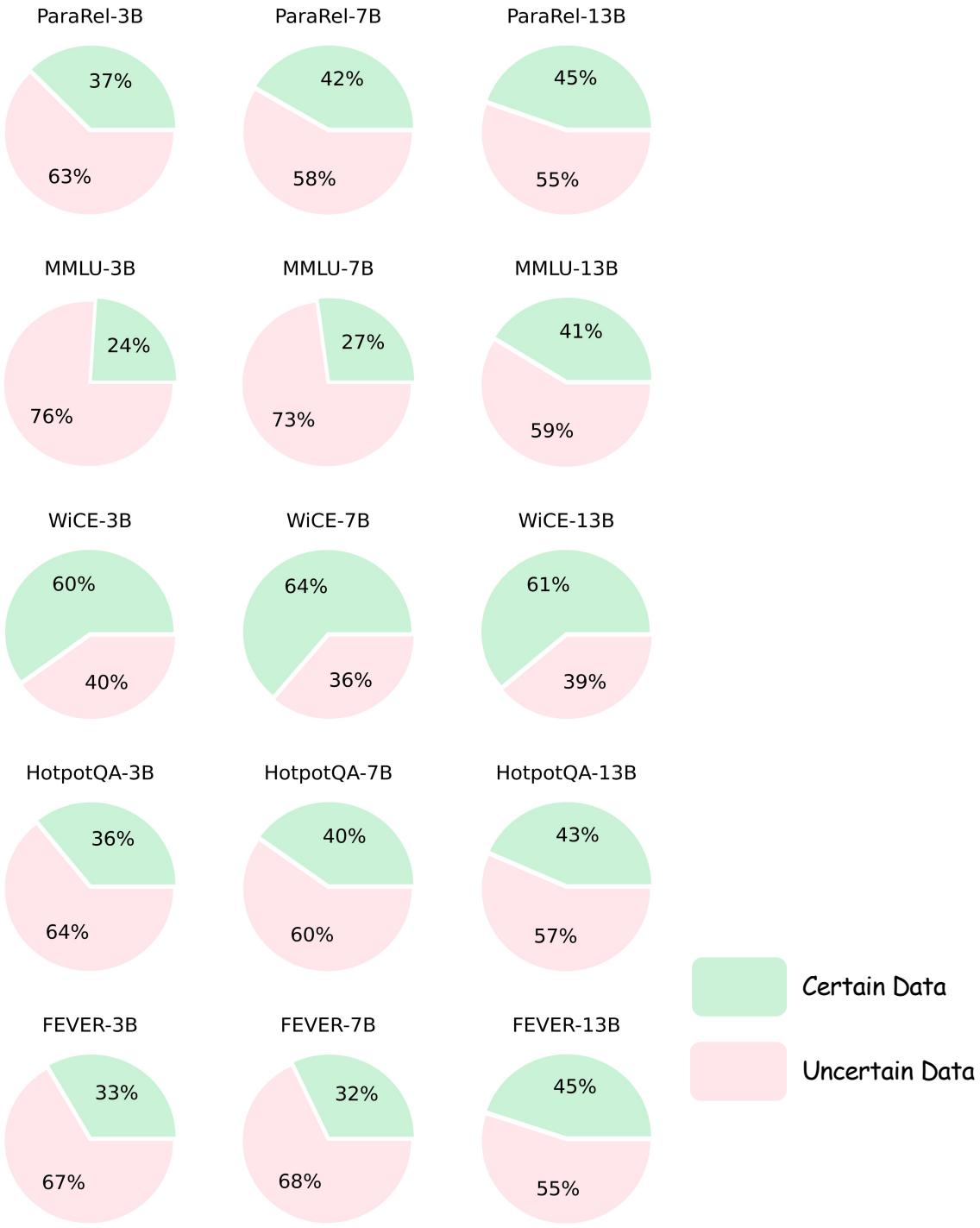


Figure 6: The data distribution of the refusal-aware datasets obtained from supervised identification strategy. The title of each sub-figure consists of the dataset name and the size of the pre-trained model used to evaluate.